Fiscal Year:	FY 2026	**Task Last Updated:**	FY 06/25/2026
PI Name:	Gur, Mert Ph.D.		
Project Title:	Probing Pathogenic Mutations Under Spaceflight-Relevant Stressors via All-Atom Molecular Dynamics Simulations		
Division Name:	Space Biology		
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Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	15260	**Congressional District:**	12
Comments:			
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No. of PhD Candidates:		**No. of Master' Degrees:**	
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COI Name (Institution):	Beheshti, Afshin Ph.D. (University of Pittsburgh) Roth, Frederick Ph.D. (University of Pittsburgh)		
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	This proposal aims to elucidate how spaceflight-relevant environmental stressors, such as altered atmospheric pressure, elevated carbon dioxide (CO#), and shifts in pH, impact the structural dynamics and functions of critical human proteins, particularly examining pathogenic genetic variants under conditions representative of extravehicular activities (EVAs) and spacecraft habitats. High-resolution studies examining these factors at all-atom spatial and femtosecond time resolution are scarce, yet this detail is essential for uncovering biomolecular mechanisms that govern disease progression in space environments. Our central objectives are twofold: first, to define how these stressors alter the conformational dynamics of Glycogen Synthase Kinase-3 beta (GSK3β), a pivotal enzyme linked to musculoskeletal health and neurological function; and second, to assess how space-like conditions influence the structural stability of the common genetic variant Methylenetetrahydrofolate reductase (MTHFR) A222V, which becomes pathogenic under		

Task Description:	specific stress conditions, such as low folate levels. We will employ high-resolution, all-atom molecular dynamics (MD) simulations to precisely replicate the micro-environments astronauts encounter during missions. This computational approach enables us to systematically control environmental variables such as pressure (ranging from hypobaric to hyperbaric conditions), CO# concentrations, and pH. Specifically, Aim 1 will analyze how these spaceflight conditions affect the structural and functional equilibrium of active and inactive conformations of GSK3β, which plays crucial roles in muscle atrophy, bone density, and central nervous system signaling. Recent experimental data from NASA's Open Science Data Repository (OSD-664) show that spaceflight markedly increases GSK-3β Ser9 inhibitory phosphorylation, highlighting the relevance of studying GSK3β under space-relevant stress conditions. Our preliminary simulation studies indicate pressure-dependent modulation of GSK3β dynamics, suggesting modest pressure variations experienced during EVAs can significantly influence protein function. Aim 2 will compare structural and dynamical responses of wild-type versus A222V mutant MTHFR proteins to these same stressors, evaluating whether conditions encountered in space can transform a benign genetic variant into a pathogenic risk factor. The choice of MTHFR A222V is driven by its high prevalence (~50% heterozygosity and ~10% homozygosity in the general population) and documented clinical relevance under conditions of low folate or oxidative stress, both plausible scenarios during spaceflight. Recent studies highlight mitochondrial dysfunction as a central dysregulation hub in space, potentially exacerbating the pathogenic potential of MTHFR variants. The significance of this work lies in its potential to provide molecular-level insights into how common genetic variations can alter astronaut health risks during space missions, addressing NASA's emphases on Precision Health, Genetic Diversity, and Age-Associated Diseases. This study represents an innovative, first-of-its-kind application of detailed MD simulations to space biology, enabling atomic-level understanding of protein behavior under realistic space conditions. This approach will provide methodological and foundational insights into how space conditions influence biomolecular processes. Results will offer critical insights to guide personalized astronaut health countermeasures, including precise nutritional supplementation, targeted pharmaceutical interventions, and optimized spacecraft environmental controls, thereby supporting NASA's broader objectives in astronaut health and precision medicine. All data and simulation outcomes will be disseminated through NASA's Open Science Data Repository and presented at relevant scientific conferences, maximizing accessibility and scientific utility.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New Project for FY2026
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